

Education

B.Tech. in Computer Science and Engineering

Indian Institute of Information Technology (IIIT) Kottayam, India

- **Relevant Coursework:** DevOps, Data Structures & Algorithms, Operating Systems, DBMS, Probability & Statistics, Linear Algebra, Optimization Techniques

Technical Skills

Programming & Scripting:	Python, Bash, C++, SQL
Software Engineering:	REST APIs, Microservices, Distributed Systems, Unit Testing, Integration Testing
Cloud & Infrastructure:	GCP, Azure, Kubernetes, Docker, Terraform
CI/CD & Automation:	Jenkins, GitHub Actions, Git, Maven
Linux & Systems:	Linux, System Debugging, Performance Analysis
Monitoring & Observability:	Prometheus, Grafana
Backend Engineering:	FastAPI, API Development, Model Serving
Machine Learning:	PyTorch, TensorFlow, Scikit-learn, XGBoost, Vertex AI

Open Source Experience

Apache SeaTunnel (ASF) — Open Source Contributor [\[Issue #10270\]](#)

- Resolved a critical distributed data-skew defect in the MySQL CDC connector causing checkpoint timeouts, `CheckpointException`, and task stalls during snapshot processing.
- Re-architected chunk-splitting logic using distribution metrics and dynamic sampling to handle sparse primary-key ranges and uneven data distribution.
- Improved workload balancing across parallel CDC readers through adaptive data partitioning and incremental database-query fallbacks.
- Developed unit and boundary-correctness tests using JUnit and Debezium mocks; contributed through multi-round Apache ASF code reviews.

QuantumLib / qsim (Open Source) [\[PR #1016\]](#)

GPU Systems Testing Contributor

- Designed integration tests to validate GPU simulation correctness across large-scale execution scenarios and boundary conditions.
- Verified runtime behavior while scaling simulations from 31 to 43 qubits, including 64-bit indexing and GPU kernel execution paths.
- Investigated GPU memory constraints with maintainers and strengthened automated CI validation before merge.
- Benchmarked GPU simulation results against reference outputs to validate correctness across diverse test cases.

Projects

End-to-End MLOps Platform — [Project Link](#)

Cloud-Native ML Engineering

Tech Stack: Python, Kubernetes, Terraform, MLflow, Seldon Core, Docker

- Built automated validation workflows for Kubernetes-based services using Python, Docker, and CI/CD automation.
- Developed integration tests for REST API endpoints deployed through Seldon Core and validated end-to-end model-serving workflows.
- Automated infrastructure verification with Terraform and analyzed runtime logs and Prometheus/Grafana metrics to identify deployment and service issues.

Autonomous Risk Evaluation & Stabilization System — [Project Link](#)

Python Systems & Time-Series Engineering

Tech Stack: Python, PyTorch, FastAPI, Prometheus, Grafana, LSTM

- Developed a Python-based forecasting pipeline for forex risk prediction using sequential financial indicators and LSTM models.
- Engineered automated drift detection and retraining workflows using configurable thresholds and model evaluation tracking.
- Deployed FastAPI inference services with Prometheus and Grafana monitoring for continuous inference and runtime observability.

High-Frequency LOB Prediction & Execution Backtesting Framework — [Project Link](#)

Python Data Engineering & Quantitative Systems

Tech Stack: Python, PyTorch, PyTorch Lightning, GPU

- Developed a modular Python pipeline for ingesting and preprocessing Level-2/Level-3 Limit Order Book datasets, including event normalization and dynamic volatility-aware target generation.
- Implemented reusable market-microstructure components for Order Flow Imbalance, queue imbalance, micro-price, bid-ask spread dynamics, and depth-based liquidity features.
- Built a GPU-accelerated CNN-Inception-LSTM forecasting pipeline in PyTorch Lightning with focal loss for short-horizon price-direction prediction.
- Designed an event-driven backtesting engine modeling execution latency, queue-position decay, maker-taker fees, and inventory constraints for realistic strategy evaluation.